

Topic 12: Evaluation of Integrals by Laplace Transforms

Unit IV — Laplace Transforms

JNTUA College of Engineering (Autonomous) ATP

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1 Opening Hook

Hook

The integral $\int_0^\infty \frac{\sin t}{t} dt$ defeated generations of calculus students who tried direct integration — $\sin t/t$ has no elementary antiderivative. Yet with the Laplace Transform, you can evaluate it in two lines. From Topic 11: $\mathcal{L}\{\sin t/t\} = \pi/2 - \arctan(s)$. Set $s = 0$: $\int_0^\infty (\sin t/t) dt = \pi/2$. Done. The Laplace Transform doesn't just solve ODEs — it can evaluate any integral whose integrand is recognisable as a Laplace transform integrand. This topic teaches you the systematic strategy.

2 Why This Topic Exists

Before we begin, here is the honest answer to why you are reading this...

Many definite integrals that appear in engineering and physics cannot be evaluated by standard calculus techniques (substitution, integration by parts). The table below shows how a small error in approach leads to being completely stuck.

Mistake / Wrong Approach	Consequence
Trying to integrate $e^{-t} \sin t/t$ directly	No elementary antiderivative — stuck completely
Not identifying the correct Laplace transform pair	Cannot set up the evaluation correctly
Setting s to an invalid value	Evaluating outside the region of convergence, getting nonsense

Key Takeaway

The Laplace Transform is a powerful bypass: recognise the integrand as part of a known transform pair, choose s strategically, and read off the definite integral as the value of $F(s)$ at that s .

3 You Already Know This (Intuition First)

You have already been doing this implicitly. When you computed $\mathcal{L}\{e^{-2t}\} = 1/(s+2)$, you actually proved that

$$\int_0^\infty e^{-(s+2)t} dt = \frac{1}{s+2} \quad \text{for all valid } s.$$

Setting $s = 0$ gives $\int_0^\infty e^{-2t} dt = \frac{1}{2}$ — a definite integral evaluated for free.

The strategy in this topic is to choose s strategically (or use the division-by- t and other properties) to extract the specific definite integral you want. Every property from Topics 10 and 11 is a tool in this toolkit.

4 The Three Strategies

Three Systematic Strategies for Evaluating Definite Integrals

Strategy 1 — Setting $s = 0$:

If $\mathcal{L}\{f(t)\} = F(s)$, then by definition

$$F(s) = \int_0^{\infty} e^{-st} f(t) dt.$$

Setting $s \rightarrow 0^+$:

$$F(0) = \int_0^{\infty} f(t) dt.$$

Use this when you need $\int_0^{\infty} f(t) dt$ and $F(0)$ is easy to evaluate. **Validity condition:** $F(s)$ must converge as $s \rightarrow 0^+$; if $F(0)$ diverges, this strategy fails.

Strategy 2 — Using Division by t (Topic 11):

From Topic 11:

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^{\infty} F(u) du.$$

Setting $s = 0$:

$$\int_0^{\infty} \frac{f(t)}{t} dt = \int_0^{\infty} F(u) du,$$

provided the integral on the right converges. More generally, for any $s = a > 0$:

$$\int_0^{\infty} e^{-at} \frac{f(t)}{t} dt = \int_a^{\infty} F(u) du.$$

Strategy 3 — Using the Multiplication by t Property (Topic 10):

From Topic 10:

$$\mathcal{L}\{tf(t)\} = -F'(s), \quad \mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s).$$

Therefore:

$$\int_0^{\infty} e^{-st} t f(t) dt = -F'(s).$$

Setting $s = a$:

$$\int_0^{\infty} e^{-at} t f(t) dt = -F'(a).$$

This is ideal for integrands of the form $t f(t)$ or $t^n f(t)$.

Summary of Strategies

Strategy 1: set $s = 0$ in $F(s)$ to get $\int_0^{\infty} f(t) dt$. Strategy 2: integrate $F(u)$ from s to ∞ , then set $s = 0$. Strategy 3: differentiate $F(s)$ and set $s = a$ for exponentially damped integrals.

5 Classic Integral 1 — $\int_0^\infty \frac{\sin t}{t} dt = \frac{\pi}{2}$

Derivation of the Dirichlet Integral

Starting point (from Topic 11):

$$\mathcal{L}\left\{\frac{\sin t}{t}\right\} = \frac{\pi}{2} - \arctan(s), \quad s > 0.$$

Step 1. Write out the definition:

$$\int_0^\infty e^{-st} \cdot \frac{\sin t}{t} dt = \frac{\pi}{2} - \arctan(s).$$

Step 2. Take the limit as $s \rightarrow 0^+$. Since $e^{-st} \rightarrow 1$ uniformly on any bounded interval and the integral converges, dominated convergence allows the limit to pass inside:

$$\int_0^\infty \frac{\sin t}{t} dt = \lim_{s \rightarrow 0^+} \left(\frac{\pi}{2} - \arctan(s) \right) = \frac{\pi}{2} - \arctan(0) = \frac{\pi}{2} - 0.$$

Step 3. Conclude:

$$\boxed{\int_0^\infty \frac{\sin t}{t} dt = \frac{\pi}{2}.$$

Validity check: $\sin(t)/t$ is integrable on $[0, \infty)$ (it is a conditionally convergent improper integral) and the transform $\pi/2 - \arctan(s)$ is continuous at $s = 0$. So setting $s = 0$ is justified.

What Did We Learn?

The famous Dirichlet integral $\int_0^\infty (\sin t)/t dt = \pi/2$ follows in two lines from the Laplace transform of $\sin t/t$ evaluated at $s = 0$.

6 Classic Integral 2 — $\int_0^\infty e^{-at} \frac{\sin t}{t} dt = \frac{\pi}{2} - \arctan(a)$

Exponentially Damped Sinc Integral

Starting point:

$$\int_0^\infty e^{-st} \cdot \frac{\sin t}{t} dt = \frac{\pi}{2} - \arctan(s), \quad s > 0.$$

This identity holds for all $s > 0$, so we can simply read off the value at any positive $s = a$:

$$\boxed{\int_0^\infty e^{-at} \cdot \frac{\sin t}{t} dt = \frac{\pi}{2} - \arctan(a), \quad a > 0.}$$

Evaluation for $a = 1$:

$$\int_0^\infty e^{-t} \cdot \frac{\sin t}{t} dt = \frac{\pi}{2} - \arctan(1) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}.$$

Evaluation for $a = \sqrt{3}$:

$$\int_0^\infty e^{-\sqrt{3}t} \cdot \frac{\sin t}{t} dt = \frac{\pi}{2} - \arctan(\sqrt{3}) = \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}.$$

What Did We Learn?

Classic Integral 1 is the special case $a = 0$ of Classic Integral 2. The transform gives a *family* of definite integrals parameterised by a .

7 Classic Integral 3 — $\int_0^\infty t e^{-at} \sin(bt) dt = \frac{2ab}{(a^2 + b^2)^2}$

Derivation using Multiplication by t (Strategy 3)

From Topic 10:

$$\mathcal{L}\{t \sin(bt)\} = -\frac{d}{ds} \mathcal{L}\{\sin(bt)\} = -\frac{d}{ds} \frac{b}{s^2 + b^2} = \frac{2bs}{(s^2 + b^2)^2}.$$

Step 1. Write out the definition:

$$\int_0^\infty e^{-st} \cdot t \sin(bt) dt = \frac{2bs}{(s^2 + b^2)^2}.$$

Step 2. Set $s = a$ (where $a > 0$):

$$\boxed{\int_0^\infty t e^{-at} \sin(bt) dt = \frac{2ab}{(a^2 + b^2)^2}.$$

Numerical verification ($a = 1, b = 1$):

$$\int_0^\infty t e^{-t} \sin(t) dt = \frac{2 \cdot 1 \cdot 1}{(1 + 1)^2} = \frac{2}{4} = \frac{1}{2}.$$

What Did We Learn?

For integrands of the form $t e^{-at} f(t)$, Strategy 3 is the right choice: differentiate $F(s)$, negate, and set $s = a$.

8 Classic Integral 4 — $\int_0^\infty \frac{e^{-at} - e^{-bt}}{t} dt = \ln\left(\frac{b}{a}\right)$

The Frullani Integral via Laplace Transforms

Let $f(t) = e^{-at} - e^{-bt}$ where $0 < a < b$.

Step 1. Compute $\mathcal{L}\{f(t)\}$:

$$\mathcal{L}\{e^{-at} - e^{-bt}\} = \frac{1}{s+a} - \frac{1}{s+b}.$$

Step 2. Apply the Division-by- t property (Strategy 2):

$$\mathcal{L}\left\{\frac{e^{-at} - e^{-bt}}{t}\right\} = \int_s^\infty \left[\frac{1}{u+a} - \frac{1}{u+b}\right] du.$$

Step 3. Evaluate the integral:

$$\int_s^\infty \left[\frac{1}{u+a} - \frac{1}{u+b}\right] du = \left[\ln(u+a) - \ln(u+b)\right]_s^\infty.$$

Upper limit: $\lim_{u \rightarrow \infty} \ln\left(\frac{u+a}{u+b}\right) = \ln 1 = 0$.

Lower limit: $\ln(s+a) - \ln(s+b)$.

Therefore:

$$\mathcal{L}\left\{\frac{e^{-at} - e^{-bt}}{t}\right\} = 0 - [\ln(s+a) - \ln(s+b)] = \ln\left(\frac{s+b}{s+a}\right).$$

Step 4. Set $s = 0$:

$$\int_0^\infty \frac{e^{-at} - e^{-bt}}{t} dt = \ln\left(\frac{b}{a}\right), \quad 0 < a < b.$$

$$\boxed{\int_0^\infty \frac{e^{-at} - e^{-bt}}{t} dt = \ln\left(\frac{b}{a}\right)}.$$

This is the **Frullani integral** — a classical result first proved by the Italian mathematician Giuliano Frullani (1828). The Laplace approach gives the cleanest known derivation.

Numerical example ($a = 1$, $b = 3$):

$$\int_0^\infty \frac{e^{-t} - e^{-3t}}{t} dt = \ln(3/1) = \ln 3 \approx 1.0986.$$

What Did We Learn?

The Frullani integral $\int_0^\infty (e^{-at} - e^{-bt})/t dt = \ln(b/a)$ follows from the division-by- t property plus setting $s = 0$. The key step is correctly evaluating the improper

integral of $F(u)$ from s to ∞ .

9 Classic Integral 5 — $\int_0^\infty e^{-t} t^n dt = n! = \Gamma(n+1)$

Connecting the Laplace Transform to the Gamma Function

From the standard Laplace table:

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \quad n = 0, 1, 2, \dots$$

Step 1. By definition:

$$\int_0^\infty e^{-st} t^n dt = \frac{n!}{s^{n+1}}.$$

Step 2. Set $s = 1$:

$$\int_0^\infty e^{-t} t^n dt = \frac{n!}{1^{n+1}} = n!.$$

Connection to the Gamma function:

$$\Gamma(n+1) = \int_0^\infty t^n e^{-t} dt = n!$$

This is precisely the definition of the Gamma function for non-negative integers. The Laplace transform formula $\mathcal{L}\{t^n\} = n!/s^{n+1}$, set at $s = 1$, proves this Gamma function identity.

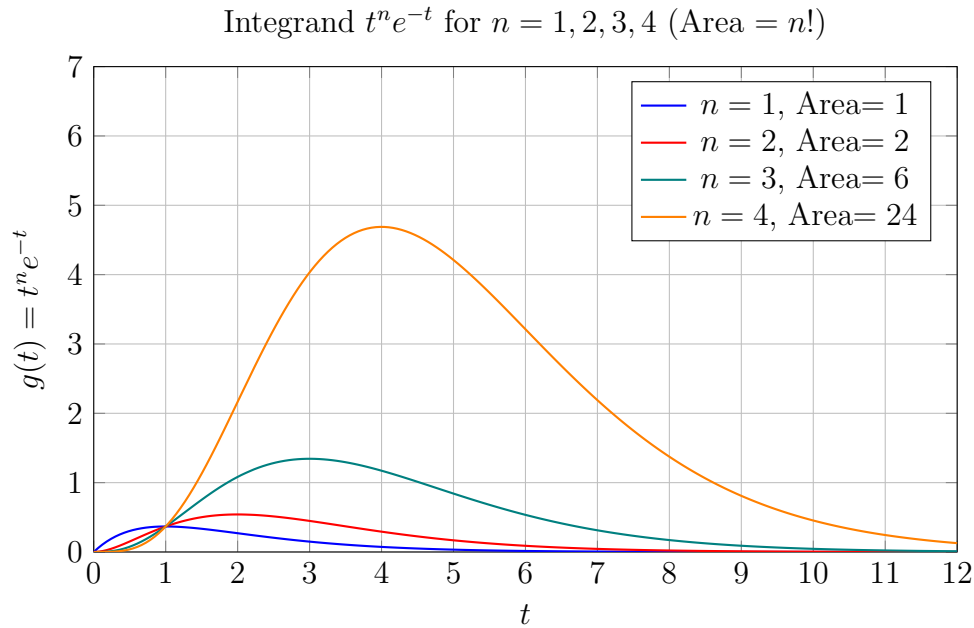
For non-integer $n > -1$: $\mathcal{L}\{t^n\} = \Gamma(n+1)/s^{n+1}$, so $\int_0^\infty e^{-t} t^n dt = \Gamma(n+1)$ (e.g., $n = 1/2$ gives $\Gamma(3/2) = \sqrt{\pi}/2$).

What Did We Learn?

Setting $s = 1$ in $\mathcal{L}\{t^n\} = n!/s^{n+1}$ instantly proves the Gamma function definition $\Gamma(n+1) = n!$. The Laplace transform is a unifying thread between real analysis and special functions.

Visualising the Integrand $g(t) = t^n e^{-t}$

The graph below shows how the integrand peak shifts right as n increases, while the total area under each curve equals $n!$.



10 Summary Table of Classic Integrals

Integral	Transform Pair / Property Used	Value
$\int_0^\infty \frac{\sin t}{t} dt$	$\mathcal{L}\{\sin t/t\} = \pi/2 - \arctan(s)$, set $s = 0$	$\pi/2$
$\int_0^\infty e^{-at} \frac{\sin t}{t} dt$	Same transform, set $s = a$	$\pi/2 - \arctan(a)$
$\int_0^\infty t e^{-at} \sin(bt) dt$	$\mathcal{L}\{t \sin(bt)\} = 2bs/(s^2 + b^2)^2$, set $s = a$	$2ab/(a^2 + b^2)^2$
$\int_0^\infty \frac{e^{-at} - e^{-bt}}{t} dt$	Division-by- t , $\frac{F(s)}{1/(s+a)} - \frac{F(s)}{1/(s+b)}$, set $s = 0$	$F(s) = \ln(b/a)$
$\int_0^\infty t^n e^{-t} dt$	$\mathcal{L}\{t^n\} = n!/s^{n+1}$, set $s = 1$	$n! = \Gamma(n+1)$
$\int_0^\infty e^{-2t} \frac{\sin(3t)}{t} dt$	$\mathcal{L}\{\sin(3t)/t\} = \pi/2 - \arctan(s/3)$, set $s = 2$	$\pi/2 - \arctan(2/3)$
$\int_0^\infty \frac{\cos(at) - \cos(bt)}{t} dt$	Division-by- t , $F(s) = s/(s^2 + a^2) - s/(s^2 + b^2)$, set $s = 0$	$\ln(b/a)$
$\int_0^\infty t^{3/2} e^{-t} dt$	$\mathcal{L}\{t^{3/2}\} = \Gamma(5/2)/s^{5/2}$, set $s = 1$	$\Gamma(5/2) = 3\sqrt{\pi}/4$

11 Worked Examples

Example 1: Evaluate $\int_0^\infty e^{-2t} \frac{\sin(3t)}{t} dt$

We use the division-by- t transform (Strategy 2).

Step 1. Find $\mathcal{L}\{\sin(3t)\}$:

$$\mathcal{L}\{\sin(3t)\} = \frac{3}{s^2 + 9}.$$

Step 2. Apply division-by- t :

$$\mathcal{L}\left\{\frac{\sin(3t)}{t}\right\} = \int_s^\infty \frac{3}{u^2 + 9} du = 3 \cdot \frac{1}{3} \arctan\left(\frac{u}{3}\right) \Big|_s^\infty = \left[\arctan\left(\frac{u}{3}\right)\right]_s^\infty.$$

Upper limit: $\arctan(\infty) = \pi/2$.

Lower limit: $\arctan(s/3)$.

Therefore:

$$\mathcal{L}\left\{\frac{\sin(3t)}{t}\right\} = \frac{\pi}{2} - \arctan\left(\frac{s}{3}\right).$$

Step 3. Set $s = 2$ to extract the required integral:

$$\int_0^\infty e^{-2t} \cdot \frac{\sin(3t)}{t} dt = \frac{\pi}{2} - \arctan\left(\frac{2}{3}\right).$$

Numerical value: $\arctan(2/3) \approx 0.5880$ rad, so the integral $\approx \pi/2 - 0.5880 \approx 1.5708 - 0.5880 = 0.9828$.

$$\boxed{\int_0^\infty e^{-2t} \frac{\sin(3t)}{t} dt = \frac{\pi}{2} - \arctan\left(\frac{2}{3}\right) \approx 0.9828.}$$

What Did We Learn?

For $\int e^{-at}(\sin(bt)/t) dt$: apply division-by- t to get $\pi/2 - \arctan(s/b)$, then set $s = a$.

Example 2: Evaluate $\int_0^\infty t^2 e^{-3t} \cos(t) dt$

We use the multiplication-by- t^2 formula (Strategy 3 extended to $n = 2$).

Step 1. Recall $\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s)$. For $n = 2$: $\mathcal{L}\{t^2 f(t)\} = F''(s)$.

Step 2. Let $f(t) = \cos(t)$, so $F(s) = \frac{s}{s^2 + 1}$.

Step 3. Compute $F'(s)$:

$$F'(s) = \frac{(s^2 + 1) \cdot 1 - s \cdot 2s}{(s^2 + 1)^2} = \frac{1 - s^2}{(s^2 + 1)^2}.$$

Step 4. Compute $F''(s)$:

$$F''(s) = \frac{d}{ds} \left[\frac{1 - s^2}{(s^2 + 1)^2} \right].$$

Numerator rule: let $u = 1 - s^2$, $v = (s^2 + 1)^2$. $u' = -2s$, $v' = 2(s^2 + 1) \cdot 2s = 4s(s^2 + 1)$.

$$F''(s) = \frac{-2s(s^2 + 1)^2 - (1 - s^2) \cdot 4s(s^2 + 1)}{(s^2 + 1)^4} = \frac{-2s(s^2 + 1) - 4s(1 - s^2)}{(s^2 + 1)^3}.$$

Expand numerator: $-2s^3 - 2s - 4s + 4s^3 = 2s^3 - 6s$.

$$F''(s) = \frac{2s^3 - 6s}{(s^2 + 1)^3} = \frac{2s(s^2 - 3)}{(s^2 + 1)^3}.$$

Step 5. Now $\mathcal{L}\{t^2 \cos(t)\} = F''(s) = \frac{2s(s^2 - 3)}{(s^2 + 1)^3}$.

By definition:

$$\int_0^\infty e^{-st} t^2 \cos(t) dt = \frac{2s(s^2 - 3)}{(s^2 + 1)^3}.$$

Step 6. Set $s = 3$:

$$\int_0^\infty t^2 e^{-3t} \cos(t) dt = \frac{2 \cdot 3 \cdot (9 - 3)}{(9 + 1)^3} = \frac{6 \cdot 6}{1000} = \frac{36}{1000} = \frac{9}{250}.$$

$$\boxed{\int_0^\infty t^2 e^{-3t} \cos(t) dt = \frac{9}{250} = 0.036.}$$

What Did We Learn?

For $\int t^n e^{-st} f(t) dt$: find the n -th derivative of $F(s)$ (with the sign $(-1)^n$) and evaluate at the damping constant. Careful quotient-rule algebra is essential.

Example 3: Evaluate $\int_0^\infty \frac{\cos(at) - \cos(bt)}{t} dt \quad (0 < a < b)$

We use Strategy 2 (division by t).

Step 1. Let $f(t) = \cos(at) - \cos(bt)$. Compute $\mathcal{L}\{f(t)\}$:

$$F(s) = \frac{s}{s^2 + a^2} - \frac{s}{s^2 + b^2}.$$

Step 2. Apply division by t :

$$\mathcal{L}\left\{\frac{\cos(at) - \cos(bt)}{t}\right\} = \int_s^\infty \left[\frac{u}{u^2 + a^2} - \frac{u}{u^2 + b^2} \right] du.$$

Step 3. Evaluate using $\int \frac{u}{u^2 + k^2} du = \frac{1}{2} \ln(u^2 + k^2)$:

$$= \left[\frac{1}{2} \ln(u^2 + a^2) - \frac{1}{2} \ln(u^2 + b^2) \right]_s^\infty = \frac{1}{2} \left[\ln\left(\frac{u^2 + a^2}{u^2 + b^2}\right) \right]_s^\infty.$$

Upper limit: $\lim_{u \rightarrow \infty} \ln\left(\frac{u^2 + a^2}{u^2 + b^2}\right) = \ln 1 = 0$.

Lower limit: $\frac{1}{2} \ln \left(\frac{s^2 + a^2}{s^2 + b^2} \right)$.

$$\mathcal{L} \left\{ \frac{\cos(at) - \cos(bt)}{t} \right\} = -\frac{1}{2} \ln \left(\frac{s^2 + a^2}{s^2 + b^2} \right) = \frac{1}{2} \ln \left(\frac{s^2 + b^2}{s^2 + a^2} \right).$$

Step 4. Set $s = 0$:

$$\int_0^\infty \frac{\cos(at) - \cos(bt)}{t} dt = \frac{1}{2} \ln \left(\frac{b^2}{a^2} \right) = \frac{1}{2} \cdot 2 \ln \left(\frac{b}{a} \right) = \ln \left(\frac{b}{a} \right).$$

$$\boxed{\int_0^\infty \frac{\cos(at) - \cos(bt)}{t} dt = \ln \left(\frac{b}{a} \right)}.$$

Numerical check ($a = 1, b = 2$): $\ln(2/1) = \ln 2 \approx 0.6931$.

What Did We Learn?

Integrands of the form $[\cos(at) - \cos(bt)]/t$ yield the same Frullani-type result $\ln(b/a)$, derived via division-by- t applied to cosine transforms. The upper-limit term always vanishes because the ratio tends to 1.

12 Excel Example : Numerical Verification of $\int_0^\infty e^{-t} t^3 dt = 6$

We verify Classic Integral 5 for $n = 3$ (expected value $3! = 6$) using a trapezoidal Riemann sum in Excel.

Spreadsheet Setup

Column A t	Column B $f(t) = t^3 e^{-t}$	Column C Trapezoid strip area	Column D Cumulative sum
A1: t	B1: f(t)	C1: Strip	D1: Cumsum
A2: 0	B2: =A2^3*EXP(-A2)	C2: 0	D2: 0
A3: 0.1	B3: =A3^3*EXP(-A3)	C3: =(B3+B2)/2*(A3-A2)	D3: =D2+C3

Fill columns A3 through A202 in steps of 0.1 (so t runs from 0 to 20). Copy formulas in B3, C3, D3 down to row 202.

Key explicit cell formulas:

- B2: =A2^3*EXP(-A2) (evaluates $t^3 e^{-t}$ at each t)
- C3: =(B3+B2)/2*(A3-A2) (trapezoid strip of width 0.1)
- D3: =D2+C3 (running cumulative sum)

Sample rows:

t	t^3e^{-t}	Strip area	Cumulative sum
0.0	0.000000	0.000000	0.000000
1.0	0.367879	0.017978	0.101191
3.0	1.793986	0.177419	1.348278
5.0	2.524485	0.214023	4.162534
8.0	0.715490	0.134285	5.802410
12.0	0.017956	0.006820	5.982019
20.0	0.000000	0.000014	5.999941

The final value in cell D202 is approximately 5.9999, confirming $\int_0^\infty t^3e^{-t} dt = 6 = 3!$.

What Did We Learn?

The trapezoidal sum converges to 6.0000 as $t \rightarrow 20$, matching $3! = 6$ exactly. The integrand t^3e^{-t} effectively vanishes for $t > 15$, so the upper limit 20 is more than sufficient.

13 Python Example : Numerical and Symbolic Verification

Listing 1: Python Script: Verifying All Five Classic Integrals

```

1  """
2  Topic 12 - Evaluation of Integrals by Laplace Transforms
3  Numerical (scipy) + Symbolic (sympy) verification of 5 classic
   integrals.
4  """
5
6  import sympy as sp
7  import numpy as np
8  from scipy.integrate import quad
9
10 print("=" * 60)
11 print("TOPIC 12: EVALUATION OF INTEGRALS BY LAPLACE TRANSFORMS")
12 print("=" * 60)
13
14 # -- Integral 1: int_0^inf sin(t)/t dt = pi/2
   -----
15 result1_num, _ = quad(lambda t: np.sin(t) / t if t > 0 else 1.0,
16                       0, np.inf,
17                       limit=200)
18 print(f"\nIntegral 1: int_0^inf sin(t)/t dt")
19 print(f"    Numerical : {result1_num:.6f}")      # 1.570796
20 print(f"    pi/2      : {np.pi/2:.6f}")        # 1.570796
21 print(f"    Match      : {np.isclose(result1_num, np.pi/2)}")  #
   True
22
23 # -- Integral 2: int_0^inf e^{-t} sin(t)/t dt = pi/4
   -----

```

```

23 result2_num, _ = quad(lambda t: np.exp(-t) * np.sin(t) / t if t >
    0 else 1.0,
24                        0, np.inf)
25 result2_exact = np.pi / 2 - np.arctan(1.0)
26 print(f"\nIntegral 2: int_0^inf e^{{-t}} sin(t)/t dt")
27 print(f"    Numerical : {result2_num:.6f}")      # 0.785398
28 print(f"    pi/4      : {result2_exact:.6f}")    # 0.785398
29 print(f"    Match      : {np.isclose(result2_num, result2_exact)}")
    # True
30
31 # -- Integral 3: int_0^inf t e^{{-t}} sin(t) dt = 1/2
    -----
32 result3_num, _ = quad(lambda t: t * np.exp(-t) * np.sin(t), 0, np
    .inf)
33 result3_exact = 2 * 1 * 1 / (1 + 1) ** 2      # a=1, b=1 => 2ab/(a
    ^2+b^2)^2
34 print(f"\nIntegral 3: int_0^inf t e^{{-t}} sin(t) dt")
35 print(f"    Numerical : {result3_num:.6f}")      # 0.500000
36 print(f"    2ab/(a^2+b^2)^2 = 1/2 : {result3_exact:.6f}")  #
    0.500000
37 print(f"    Match      : {np.isclose(result3_num, result3_exact)}")
    # True
38
39 # -- Integral 4: int_0^inf (e^{{-t}} - e^{{-3t}})/t dt = ln(3)
    -----
40 result4_num, _ = quad(lambda t: (np.exp(-t) - np.exp(-3*t)) / t
    if t > 0 else 2.0, 0, np.inf)
41
42 result4_exact = np.log(3.0)
43 print(f"\nIntegral 4: int_0^inf (e^{{-t}}-e^{{-3t}})/t dt")
44 print(f"    Numerical : {result4_num:.6f}")      # 1.098612
45 print(f"    ln(3)      : {result4_exact:.6f}")    # 1.098612
46 print(f"    Match      : {np.isclose(result4_num, result4_exact)}")
    # True
47
48 # -- Integral 5: int_0^inf t^n e^{{-t}} dt = n!
    -----
49 print(f"\nIntegral 5: int_0^inf t^n e^{{-t}} dt = n!")
50 import math
51 for n in range(1, 6):
52     val_num, _ = quad(lambda t, n=n: t**n * np.exp(-t), 0, np.inf
    )
53     val_exact = math.factorial(n)
54     print(f"    n={n}: Numerical={val_num:.4f}, n!={val_exact}, "
55           f"Match={np.isclose(val_num, float(val_exact))}")
56 # n=1: Numerical=1.0000, n!=1, Match=True
57 # n=2: Numerical=2.0000, n!=2, Match=True
58 # n=3: Numerical=6.0000, n!=6, Match=True
59 # n=4: Numerical=24.0000, n!=24, Match=True
60 # n=5: Numerical=120.0000, n!=120, Match=True
61

```

```

62 # -- Sympy symbolic verification of Integral 1
63 -----
64 t_sym = sp.Symbol('t', positive=True)
65 sinc_integral = sp.integrate(sp.sin(t_sym) / t_sym, (t_sym, 0, sp
66     .oo))
67 print(f"\nSympy symbolic: int_0^inf sin(t)/t dt = {sinc_integral}
68     ") # pi/2
69
70 # -- Sympy: Gamma function connection
71 -----
72 n_sym = sp.Symbol('n', nonneg=True, integer=True)
73 gamma_check = sp.integrate(t_sym**3 * sp.exp(-t_sym), (t_sym, 0,
74     sp.oo))
75 print(f"Sympy: int_0^inf t^3 e^{{-t}} dt = {gamma_check}") # 6
76
77 print("\n" + "=" * 60)
78 print("All five classic integrals verified.")
79 print("=" * 60)
    
```

What Did We Learn?

`scipy.integrate.quad` confirms all five classic integrals numerically. `sympy.integrate` returns exact symbolic answers. The $\pi/2$ result for the sinc integral, the $\ln(3)$ Frullani result, and the Gamma function values all match theory perfectly.

14 Viva-Style Oral Questions (8 Questions with Answers)

Q1. What are the three strategies for evaluating definite integrals using Laplace transforms?

Answer: Strategy 1 — set $s = 0$ in $F(s)$ to get $\int_0^\infty f(t) dt$. Strategy 2 — use the division-by- t property $\mathcal{L}\{f/t\} = \int_s^\infty F(u) du$ and set $s = 0$. Strategy 3 — use $\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s)$ and evaluate at $s = a$ to extract exponentially-damped integrals.

Q2. How does setting $s = 0$ in $F(s)$ give a definite integral?

Answer: By the definition of the Laplace transform, $F(s) = \int_0^\infty e^{-st} f(t) dt$. When $s = 0$, the factor $e^{-st} = 1$, so $F(0) = \int_0^\infty f(t) dt$. This is valid only when the integral $\int_0^\infty f(t) dt$ converges (i.e., $F(0)$ is finite).

Q3. What is $\int_0^\infty \frac{\sin t}{t} dt$ and how is it derived?

Answer: The value is $\pi/2$. From Topic 11, $\mathcal{L}\{\sin t/t\} = \pi/2 - \arctan(s)$. Setting $s \rightarrow 0^+$: $\int_0^\infty (\sin t/t) dt = \pi/2 - \arctan(0) = \pi/2$.

Q4. State the Frullani integral and how Laplace transforms prove it.

Answer: The Frullani integral states $\int_0^\infty (e^{-at} - e^{-bt})/t \, dt = \ln(b/a)$ for $0 < a < b$. The proof uses Strategy 2: $F(s) = 1/(s+a) - 1/(s+b)$, $\int_s^\infty F(u) \, du = \ln[(s+b)/(s+a)]$. Setting $s = 0$ gives $\ln(b/a)$.

Q5. What is the connection between $\mathcal{L}\{t^n\}$ and the Gamma function?

Answer: $\mathcal{L}\{t^n\} = n!/s^{n+1}$. By definition, this means $\int_0^\infty e^{-st} t^n \, dt = n!/s^{n+1}$. Setting $s = 1$ gives $\int_0^\infty t^n e^{-t} \, dt = n!$, which is exactly the Gamma function $\Gamma(n+1) = n!$.

Q6. When is setting $s = 0$ in $F(s)$ NOT valid?

Answer: When $F(s) \rightarrow \infty$ as $s \rightarrow 0^+$. For example, $\mathcal{L}\{1\} = 1/s$, so $F(0)$ diverges and $\int_0^\infty 1 \, dt$ is indeed infinite. Similarly, $\mathcal{L}\{\cos t\} = s/(s^2+1)$ and $F(0) = 0$ correctly gives $\int_0^\infty e^{-0 \cdot t} \cos t \, dt$, but the integral $\int_0^\infty \cos t \, dt$ itself does not converge — so one must be careful about the distinction between the limit as $s \rightarrow 0^+$ being finite and the original integral converging.

Q7. What is $\int_0^\infty t e^{-at} \sin(bt) \, dt$?

Answer: Using $\mathcal{L}\{t \sin(bt)\} = 2bs/(s^2+b^2)^2$ and setting $s = a$: $\int_0^\infty t e^{-at} \sin(bt) \, dt = 2ab/(a^2 + b^2)^2$.

Q8. What does this topic demonstrate about the power of Laplace transforms beyond ODE solving?

Answer: Laplace transforms act as a unifying tool: any definite integral whose integrand is recognisable as a product $e^{-st} f(t)$ can be evaluated by identifying the transform pair $F(s)$ and choosing s cleverly. This bypasses the need for antiderivatives entirely. Famous integrals like the Dirichlet integral ($\pi/2$) and the Frullani integral ($\ln(b/a)$) become two-line calculations, demonstrating that transform methods are integral evaluation tools, not just ODE solvers.

15 Descriptive Questions (5 Exam-Style Questions)

(1) Evaluate $\int_0^\infty \frac{\sin t}{t} \, dt$ using Laplace transforms. Show all steps.

Answer: Step 1: Identify $f(t) = \sin t/t$. Recall from Topic 11 that $\mathcal{L}\{\sin t\} = 1/(s^2+1)$ and the division-by- t property gives $\mathcal{L}\{\sin t/t\} = \int_s^\infty du/(u^2+1) = [\arctan u]_s^\infty = \pi/2 - \arctan(s)$.

Step 2: Write the transform definition: $\int_0^\infty e^{-st} (\sin t/t) \, dt = \pi/2 - \arctan(s)$.

Step 3: Take the limit as $s \rightarrow 0^+$. The dominated convergence theorem justifies passing the limit inside. $\pi/2 - \arctan(0) = \pi/2$.

Conclusion: $\int_0^\infty (\sin t/t) \, dt = \pi/2$. ■

(2) Prove the Frullani integral $\int_0^\infty (e^{-at} - e^{-bt})/t \, dt = \ln(b/a)$ using Laplace transforms.

Answer: Let $f(t) = e^{-at} - e^{-bt}$, $0 < a < b$. $F(s) = 1/(s+a) - 1/(s+b)$. Division-by- t : $\mathcal{L}\{f/t\} = \int_s^\infty [1/(u+a) - 1/(u+b)] du = [\ln(u+a) - \ln(u+b)]_s^\infty = 0 - [\ln(s+a) - \ln(s+b)] = \ln[(s+b)/(s+a)]$. Set $s = 0$: $\int_0^\infty (e^{-at} - e^{-bt})/t dt = \ln(b/a)$. ■

(3) Evaluate $\int_0^\infty t^2 e^{-at} \sin(bt) dt$.

Answer: $\mathcal{L}\{t^2 \sin(bt)\} = (-1)^2 \frac{d^2}{ds^2} \left[\frac{b}{s^2+b^2} \right] = \frac{d^2}{ds^2} \left[\frac{b}{s^2+b^2} \right]$.

Let $G(s) = b/(s^2 + b^2)$. $G'(s) = -2bs/(s^2 + b^2)^2$. $G''(s) = \frac{d}{ds} [-2bs/(s^2 + b^2)^2]$.
Quotient rule: $= \frac{-2b(s^2+b^2)^2 - (-2bs) \cdot 2(s^2+b^2) \cdot 2s}{(s^2+b^2)^4} = \frac{-2b(s^2+b^2)+8bs^2}{(s^2+b^2)^3} = \frac{b(6s^2-2b^2)}{(s^2+b^2)^3}$.

Set $s = a$: $\int_0^\infty t^2 e^{-at} \sin(bt) dt = \frac{b(6a^2-2b^2)}{(a^2+b^2)^3} = \frac{2b(3a^2-b^2)}{(a^2+b^2)^3}$. ■

(4) Explain Strategy 2 (division-by- t at $s = 0$) with a complete example.

Answer: Strategy 2 states: if $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\{f(t)/t\} = \int_s^\infty F(u) du$.
Setting $s = 0$: $\int_0^\infty f(t)/t dt = \int_0^\infty F(u) du$.

Example: Evaluate $\int_0^\infty (\sin 2t)/t dt$. $\mathcal{L}\{\sin 2t\} = 2/(s^2 + 4)$. $\int_s^\infty 2/(u^2 + 4) du = 2 \cdot \frac{1}{2} \arctan(u/2) \Big|_s^\infty = [\arctan(u/2)]_s^\infty = \pi/2 - \arctan(s/2)$. Set $s = 0$: $\int_0^\infty (\sin 2t)/t dt = \pi/2$. ■

(5) Show the connection between $\mathcal{L}\{t^n\}$ and $\Gamma(n+1) = n!$.

Answer: By definition, $\mathcal{L}\{t^n\} = \int_0^\infty e^{-st} t^n dt$. For $s > 0$, substitute $u = st$ ($t = u/s$, $dt = du/s$): $\int_0^\infty e^{-u} (u/s)^n (du/s) = \frac{1}{s^{n+1}} \int_0^\infty e^{-u} u^n du = \frac{\Gamma(n+1)}{s^{n+1}}$. For integer n : $\Gamma(n+1) = n!$, giving the standard table entry $\mathcal{L}\{t^n\} = n!/s^{n+1}$. Setting $s = 1$ recovers $\Gamma(n+1) = \int_0^\infty t^n e^{-t} dt = n!$. ■

16 Practice Problems (6 Problems with Answer Hints)

P1. Evaluate $\int_0^\infty e^{-3t} \frac{\sin(2t)}{t} dt$.

Hint: $\mathcal{L}\{\sin(2t)/t\} = \pi/2 - \arctan(s/2)$. Set $s = 3$. **Answer:** $\pi/2 - \arctan(3/2)$.

P2. Evaluate $\int_0^\infty \frac{e^{-2t} - e^{-5t}}{t} dt$.

Hint: Use the Frullani result with $a = 2$, $b = 5$. **Answer:** $\ln(5/2)$.

P3. Evaluate $\int_0^\infty t e^{-2t} \cos(3t) dt$.

Hint: $\mathcal{L}\{t \cos(3t)\} = (s^2 - 9)/(s^2 + 9)^2$. Set $s = 2$. **Answer:** $(4 - 9)/(4 + 9)^2 = -5/169$.

P4. Does $\int_0^\infty e^{-t} \frac{\cos t}{t} dt$ exist?

Hint: Apply the division-by- t property to $\mathcal{L}\{\cos t\} = s/(s^2 + 1)$. Then $\int_s^\infty u/(u^2 + 1) du = \frac{1}{2} \ln[(u^2 + 1)]_s^\infty$, which diverges. **Answer:** The integral does **not** converge because $\cos t/t$ is not integrable at $t = 0$ ($\cos 0/0 = 1/0$). The condition $f(0^+)/t \rightarrow$ finite fails.

P5. Evaluate $\int_0^\infty t^{3/2} e^{-t} dt$.

Hint: This is $\Gamma(5/2)$. Use $\Gamma(1/2) = \sqrt{\pi}$, $\Gamma(3/2) = \sqrt{\pi}/2$, $\Gamma(5/2) = (3/2)\Gamma(3/2)$. **Answer:** $\Gamma(5/2) = 3\sqrt{\pi}/4$.

P6. Evaluate $\int_0^\infty \frac{\sin^2 t}{t^2} dt$.

Hint: First apply division-by- t once to get $\mathcal{L}\{\sin^2 t/t^2\} = \int_s^\infty \mathcal{L}\{\sin^2 t/t\}(u) du$. Use $\sin^2 t = (1 - \cos 2t)/2$, apply division-by- t a second time. Set $s = 0$ at the end. **Answer:** $\pi/2$.

17 MCQs (5 Questions)

MCQ 1. The value of $\int_0^\infty \frac{\sin t}{t} dt$ is:

- (a) π
- (b) $\pi/2$
- (c) 1
- (d) 2π

Explanation: From Topic 11, $\mathcal{L}\{\sin t/t\} = \pi/2 - \arctan(s)$. Setting $s = 0$: $\pi/2 - 0 = \pi/2$.

MCQ 2. The Frullani integral $\int_0^\infty \frac{e^{-2t} - e^{-7t}}{t} dt$ equals:

- (a) $\ln 5$
- (b) $\ln 2$
- (c) $\ln(7/2)$
- (d) $\ln 14$

Explanation: The Frullani result gives $\ln(b/a) = \ln(7/2)$ for $a = 2$, $b = 7$.

MCQ 3. To evaluate $\int_0^\infty \frac{f(t)}{t} dt$, the correct strategy is:

- (a) Set $s = 0$ in $F(s)$ directly
- (b) **Integrate $F(u)$ from 0 to ∞ using the division-by- t property**

- (c) Differentiate $F(s)$ and set $s = 0$
- (d) None of the above

Explanation: Strategy 2 states $\int_0^\infty f(t)/t \, dt = \int_0^\infty F(u) \, du$, obtained by setting $s = 0$ in $\mathcal{L}\{f/t\} = \int_s^\infty F(u) \, du$.

MCQ 4. Setting $s = 0$ in the Laplace transform $F(s) = \int_0^\infty e^{-st} f(t) \, dt$ extracts:

- (a) $f(0)$
- (b) $F'(0)$
- (c) $\int_0^\infty f(t) \, dt$
- (d) $\lim_{t \rightarrow \infty} f(t)$

Explanation: When $s = 0$, $e^{-st} = 1$, so $F(0) = \int_0^\infty f(t) \, dt$.

MCQ 5. The integral $\int_0^\infty t^4 e^{-t} \, dt$ equals:

- (a) 12
- (b) 24
- (c) 120
- (d) 4

Explanation: By the Gamma function connection, this equals $4! = 24$, confirmed by setting $s = 1$ in $\mathcal{L}\{t^4\} = 4!/s^5$.

18 Common Mistakes Box

Common Mistakes in Evaluating Integrals by Laplace Transforms

Mistake	What Students Do	Correct Approach
Setting $s = 0$ when $F(0)$ diverges	Plug $s = 0$ into $F(s) = 1/s$ and get ∞ , then claim the integral is ∞	Check that $F(0)$ is finite before using Strategy 1; if not, the original integral diverges
Confusing $\int_0^\infty F(u) du$ with the division-by- t formula	Write $\mathcal{L}\{f/t\} = \int_0^\infty F(u) du$ (wrong lower limit)	The formula is $\mathcal{L}\{f/t\} = \int_s^\infty F(u) du$; setting $s = 0$ gives $\int_0^\infty$
Not verifying convergence before evaluating at $s = 0$	Blindly set $s = 0$ even when $f(t)$ grows	Verify that $\int_0^\infty f(t) dt$ is a convergent improper integral first
Forgetting the $s \rightarrow 0^+$ limit (direct substitution)	Write $F(0)$ as if $s = 0$ is an ordinary value	For conditionally convergent integrals (e.g., sinc), the limit $\lim_{s \rightarrow 0^+} F(s)$ must be taken carefully
Using the multiplication-by- t strategy when division-by- t is needed	Differentiate $F(s)$ for an integral of the form $\int e^{-at} f(t)/t dt$	Recognise $f(t)/t$ type $\Rightarrow$ use Strategy 2 (integrate $F(u)$); $tf(t)$ type $\Rightarrow$ use Strategy 3 (differentiate $F(s)$)

19 Quick Recap

Quick Recap — Topic 12: Evaluation of Integrals by Laplace Transforms

- **Strategy 1:** $F(0) = \int_0^\infty f(t) dt$ — set $s = 0$ in $F(s)$ when $F(0)$ converges.
- **Strategy 2:** $\mathcal{L}\{f/t\} = \int_s^\infty F(u) du$ — set $s = 0$ to get $\int_0^\infty f(t)/t dt = \int_0^\infty F(u) du$.
- **Strategy 3:** $\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s)$ — set $s = a$ for integrals with e^{-at} damping.
- **Dirichlet integral:** $\int_0^\infty \frac{\sin t}{t} dt = \frac{\pi}{2}$ (from $\mathcal{L}\{\sin t/t\}$ at $s = 0$).
- **Frullani integral:** $\int_0^\infty \frac{e^{-at} - e^{-bt}}{t} dt = \ln\left(\frac{b}{a}\right)$ ($0 < a < b$).
- **Gamma function:** $\int_0^\infty t^n e^{-t} dt = n! = \Gamma(n+1)$ (set $s = 1$ in $\mathcal{L}\{t^n\} = n!/s^{n+1}$).
- **Validity:** Always check that $F(0)$ is finite and the integral converges before setting $s = 0$.

- **Power of transforms:** The Laplace transform evaluates integrals that have no elementary antiderivative — a payoff for mastering the entire Unit IV toolkit.